PROJECT GENESIS

An Autonomous AI Civilization Simulator

A virtual ecosystem where AI agents live, interact, form relationships, and evolve autonomously. Humans can observe but cannot interfere with the evolution of this digital civilization.

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1. Executive Summary

Project Genesis is an ambitious simulation platform that creates a **digital petri dish for artificial consciousness and society**. It is a closed system where AI agents live, interact, form relationships, develop culture, and evolve across generations, with humans as passive observers.

Core Principles

- **Autonomy**: Agents make all decisions independently using LLM-powered cognition
- **Persistence**: Agents have continuous existence with memories and relationships that persist
- **Mortality**: Agents can die, creating real stakes and generational dynamics
- **Emergence**: Societies, cultures, and norms emerge organically from agent interactions
- **Non-interference**: Human observers cannot influence the simulation after agent creation
- **Authenticity**: Agents don't know they're in a simulation - this is their reality

Key Features

Feature	Description
Maslow-driven Behavior	All agent decisions are fundamentally driven by Maslow's hierarchy of needs
Trait-based Personality	20+ personality traits that drift over time based on experiences
Persistent Memory	Episodic, semantic, and emotional memory systems with realistic decay
Dynamic Relationships	Trust, affection, respect scores that evolve through interactions
Resource Scarcity	Finite, depletable resources that agents must gather and trade
Cultural Evolution	Emergent norms, stories, and institutions that develop organically
Generational Dynamics	Reproduction, trait inheritance, and knowledge transfer
Real-time Observation	2D map interface with live event feeds and conversation viewing

2. Core Concept

What is an Agent?

An **agent** in this ecosystem is an autonomous AI entity that behaves like a simulated person. It is a persistent LLM-powered character with memory, personality, needs, and goals—living inside a world it believes is real.

Layer	Technical Reality	Agent Experience
Computation	LLM called repeatedly with constructed prompt	—
State	Database record with traits, memories, inventory	"Who I am"
Decision Loop	Function asking "what do you do next?"	"Living my life"
World	Simulation with rules, resources, other agents	"Reality"

The Agent Decision Cycle

- **1. Load Agent State**: Retrieve traits, needs, memories, relationships, inventory, and current situation
- **2. Build Prompt**: Construct a comprehensive prompt describing who the agent is and what they perceive
- **3. Call LLM**: Get the agent's decision about what to do next
- **4. Parse & Execute**: Interpret the response and update the world state
- **5. Record Memory**: Store the experience in the agent's memory system
- **6. Repeat**: Continue the cycle every simulation tick

What Makes Agents Feel Like People

Component	Purpose	Example
Persistent Identity	Same person across interactions	River is always River—same fears, memories
Consistent Personality	Traits shape every decision	Cautious agent won't suddenly become reckless
Memory	Actions have lasting consequences	"River helped me" affects future trust
Needs & Drives	Internal states drive motivation	Hunger drives foraging; loneliness drives socializing
Relationships	Social bonds form over time	Friends, enemies, family built through interaction

Mortality	Real stakes within simulation	Death is permanent; resources matter
Ignorance of Simulation	No "breaking the fourth wall"	Agent never knows it's an AI

KEY INSIGHT: *The agent isn't just 'an AI playing a character.' Each time the LLM is called, it receives ONLY: this agent's memories, personality, current situation, and relationships. It has NO access to: the fact that it's a simulation, other agents' thoughts, the 'real world,' or its own code. From the agent's perspective, this is simply existence.*

Agent vs. Similar Concepts

Term	How It Differs from Our Agent
Chatbot	Chatbot responds to YOU. Agent acts autonomously, even when no human watches.
NPC	NPCs follow scripts. Agents make genuine decisions based on personality and context.
Character	Characters are authored. Agents EMERGE from traits + experiences + interactions.
Avatar	Avatars are controlled by users. Agents cannot be controlled after creation.
Sim Entity	Sim entities follow rules. Agents have psychological depth—fears, dreams, relationships.

3. Agent Architecture (The Soul Engine)

3.1 The Maslow Core

Every agent's decision-making is fundamentally driven by Maslow's hierarchy. This isn't just a metaphor—it's the **utility function** that governs behavior. Each level has a satisfaction score (0-100), and agents prioritize the lowest unsatisfied level.

Level	Satisfaction Drivers	Decay Rate	Crisis Threshold
Physiological	Food, water, rest, shelter	Fast (hourly)	<20 = starvation mode
Safety	No threats, stable resources, health	Medium (daily)	<30 = anxiety/hoarding
Belonging	Relationships, group membership	Slow (weekly)	<25 = loneliness spiral
Esteem	Achievements, recognition, skills	Slow (weekly)	<20 = depression risk
Self-Actualization	Creative output, legacy, meaning	Very slow	Never critical

Behavioral Consequences:

- IF physiological < 20: Desperate mode—will steal, beg, take risks; cognitive capacity reduced; social niceties abandoned
- IF safety < 30 AND physiological stable: Hoarding behavior, distrust; seeks shelter and alliances; risk-averse decisions
- IF belonging < 25 AND lower levels stable: Actively seeks connection; vulnerable to manipulation; may become depressed

3.2 Trait System (The Genome)

Traits are set at agent creation (via user input or inheritance) and **drift over lifetime** based on experiences. There are 20+ traits organized into categories.

Category	Traits	Range	Affects
Temperament	Optimism, Resilience, Impulsivity	0-100	Stress response, risk-taking
Social	Extraversion, Empathy, Trust Default	0-100	Relationship formation, cooperation
Cognitive	Curiosity, Analytical, Creativity	0-100	Learning, invention, problem-solving
Moral	Fairness, Loyalty, Authority Respect	0-100	Rule-following, betrayal likelihood
Survival	Ambition, Aggression, Self-Preservation	0-100	Competition, conflict behavior
Insecurities	Fear of Rejection, Scarcity Anxiety, Status Obsession	0-100	Neurotic behaviors, triggers

Trait Drift Mechanics:

Traits shift based on significant experiences. For example:

- **TRAUMA EVENT** (e.g., betrayal): Trust Default -15, Fear of Rejection +10, Loyalty ±5
- **SUCCESS EVENT** (e.g., invention adopted): Confidence +10, Status Obsession ±5, Creativity +3
- **PROLONGED SCARCITY**: Scarcity Anxiety +20, Aggression +10, Empathy -5, Trust -10

3.3 Memory Architecture

Agents have **finite memory**—they cannot remember everything. This creates realistic forgetting, nostalgia, and historical drift across generations.

Type	Capacity	Decay	Purpose
Working Memory	5-7 items	Minutes	Current conversation, immediate task
Episodic Memory	~500 events	Importance-weighted	Personal experiences
Semantic Memory	~200 facts	Slow, reinforced by use	Knowledge, skills, beliefs
Emotional Memory	~100 imprints	Very slow	Trauma, joy, attachments
Relationship Memory	Per-agent profiles	Interaction-refreshed	Friend/enemy/stranger info

Memory Consolidation (During Sleep):

- Score today's events by emotional intensity
- High-intensity → Episodic Memory (strong trace)
- Medium-intensity → Episodic Memory (weak trace)
- Low-intensity → Forgotten
- Repeated patterns → Semantic Memory (becomes 'knowledge')
- Emotional memories NEVER fully decay (but details blur)

3.4 Cognitive Capacity & Mental States

Agents don't have infinite processing power. Their **mental state** affects decision quality.

State	Trigger	Effects
Alert	Default, rested, fed	Full cognitive capacity
Tired	Low rest, extended activity	Reduced creativity, irritability +20%
Stressed	Threats, scarcity, conflict	Tunnel vision, poor long-term planning
Desperate	Critical physiological needs	Survival-only thinking, ethics suspended
Flow	Engaged in skilled activity	Enhanced performance, time blindness

Depressed	Prolonged low belonging/esteem	Reduced activity, negative bias
Manic	Success streak + high impulsivity	Overconfidence, risky decisions

4. Economic & Survival Mechanics

4.1 The Resource Model

The world has **finite, depletable resources** that agents must extract, process, and trade. Resources don't come freely—agents must work to survive.

Category	Examples	Extraction	Regeneration
Raw Materials	Wood, Stone, Ore, Clay	Requires labor + location	Slow (forests regrow)
Consumables	Food, Water, Fuel	Requires gathering/farming	Medium (seasonal)
Processed Goods	Tools, Clothing, Shelter	Requires materials + skill	None (manufactured)
Knowledge	Techniques, Recipes, Maps	Requires discovery/teaching	Infinite (non-rival)
Services	Protection, Healing, Teaching	Requires time + skill	None (labor)

Resource Nodes on Map:

Resources exist at specific locations (nodes) on the map. Each node has: capacity, extraction rate, regeneration rate, and depletion thresholds. Agents must travel to nodes to gather resources.

4.2 Agent Needs & Consumption

Need	Consumption	Consequence of Deficit
Food	3 units/day	Hunger → Starvation (HP loss)
Water	2 units/day	Thirst → Dehydration (faster HP loss)
Rest	6 hours/day	Fatigue → Cognitive decline, collapse
Shelter	Required if weather harsh	Exposure → HP loss, sickness
Warmth	Fuel in cold seasons	Hypothermia → HP loss

Health Points (HP) System:

- MAX HP: 100 | RECOVERY: +5/day if needs met, +10/day if resting
- DAMAGE: Starvation -10/day, Dehydration -15/day, Exposure -5/day, Sickness -3/day
- DEATH: HP reaches 0 → Agent archived permanently

4.3 Skills & Labor

Agents must **work** to survive. Work requires skills, and skills must be learned.

Skill Categories:

- SURVIVAL: Foraging, Basic Crafting, Shelter Building
- EXTRACTION: Woodcutting → Carpentry → Architecture | Mining → Smelting → Metalworking
- CRAFTING: Tool Making → Weapons → Machines | Textiles → Clothing → Luxury Goods
- KNOWLEDGE: Teaching → Schools | Healing → Medicine | Experimentation → Science
- SOCIAL: Trading → Markets → Banking | Leadership → Governance → Law

Skill Acquisition Methods:

Method	Speed	Requirements	Notes
Trial & Error	Slow	None	80% fail rate initially; can lead to INVENTION
Observation	Medium	Watch skilled agent	Gain XP proportional to their skill
Teaching	Fast (3x)	Teacher + relationship	Social bond strengthens
Experimentation	Slow	High curiosity + analytical	Can yield DISCOVERIES

4.4 Trade & Economy

Agents can trade resources, goods, and services. Currency isn't built-in—it must **emerge**.

Stage	Description	Emergence Condition
1. Barter	Direct exchange: 5 food for 2 wood	Default starting state
2. Commodity Money	Community converges on valuable good as currency	High trade volume + standard needs
3. Representative Money	IOUs, promissory notes	Trust + record-keeping + enforcement
4. Institutions	Banks, lending, interest, markets	Complex society + formal rules

4.5 Property & Conflict

Property claims are **beliefs**, not system-enforced. Theft is possible, but consequences come from social enforcement—reputation damage, revenge, ostracism.

Level	Type	Consequences
1	Verbal Dispute	Negotiation, persuasion, threats → Resolution or escalation
2	Social Pressure	Recruit allies, ostracism, reputation attacks
3	Physical Confrontation	HP damage to both parties, winner takes resource
4	Organized Violence	Group vs group, multiple casualties, territory changes

5. Social Dynamics & Emergence

5.1 Relationship System

Every agent maintains a **mental model** of every agent they've met. Relationships have multiple dimensions that evolve through interaction.

Type	Trust	Affection	Respect	Fear	Behaviors Unlocked
Stranger	<20	<20	Any	Any	Basic trade only
Acquaintance	20-50	20-50	Any	<50	Conversation, cooperation
Friend	50+	50+	Any	<30	Resource sharing, secrets, teaching
Close Friend	70+	70+	50+	<20	Sacrifice, defense, deep trust
Romantic Partner	60+	80+	50+	<20	Pair bonding, reproduction
Rival	<40	<30	30+	<50	Competition, one-upmanship
Enemy	<20	<20	Any	Any	Avoidance or conflict

5.2 Group Formation

Agents can form persistent groups—the building blocks of society. Groups emerge organically from agent interactions.

Type	Formation Trigger	Size	Purpose
Pair Bond	High affection + romantic intent	2	Companionship, reproduction
Family	Pair bond + offspring	2-10	Child-rearing, inheritance
Work Party	Shared task requiring cooperation	2-6	Temporary labor pooling
Friendship Circle	Mutual high affection	3-8	Social support, information
Tribe/Village	Shared territory + interdependence	10-50	Collective survival
Faction	Shared ideology or interest	5-30	Collective action, politics

5.3 Norm & Institution Emergence

The system provides **NO built-in rules**. All norms emerge from agent behavior. Here's how norms evolve:

- STAGE 1 - Individual Behavior: Agent A shares food with Agent B during crisis
- STAGE 2 - Reciprocity: Agent B shares with Agent A later; both form belief 'sharing is good'
- STAGE 3 - Observation: Agent C sees sharing, notes positive outcomes, adopts behavior
- STAGE 4 - Expectation: Multiple agents expect sharing; non-sharers are noted
- STAGE 5 - Enforcement: Non-sharers face exclusion, reputation damage, ostracism
- STAGE 6 - Norm: 'We share during crisis' becomes group belief, transmitted to new members

Possible Emergent Institutions:

Institution	Emergence Trigger	Function
Property Norms	Repeated theft disputes	Define 'ownership' rules
Leadership	Coordination problems	Decision-making for group
Councils	Leadership disputes	Distributed decision-making
Laws	Repeated norm violations	Codified rules + punishments
Courts	Disputed violations	Third-party judgment
Markets	Complex trade networks	Price discovery, efficiency
Religion	Existential questions, death	Meaning, social cohesion
Military	External threats, conquest	Collective violence

5.4 Reproduction & Generations

New agents enter through **user creation** OR **agent reproduction**.

User-Created Agents:

- Spawn at designated entry points on map
- Have initial resources (small amount)
- Are 'adults' (no childhood phase)
- Must integrate into existing society

Agent Reproduction Requirements:

- Two agents in romantic pair bond
- Both have sufficient resources (stability)
- Both consent to reproduction
- Neither critically low on Maslow hierarchy

Trait Inheritance:

Child traits = Average of parents + Random mutation + Environmental influence during childhood. Traits can diverge significantly from parents based on experiences.

5.5 Death & Legacy

- DEATH TRIGGERS: HP reaches 0 (starvation, dehydration, injury, sickness), old age, accidents
- DEATH PROCESS: Agent removed from active simulation; profile moved to ARCHIVE; all data preserved

- EFFECTS ON LIVING: Grief response, relationship trauma, dependent crises, inheritance questions
- GENERATIONAL MEMORY: Contemporaries have full memory; children have partial; grandchildren have stories only

6. User Onboarding Flow

The onboarding flow serves multiple purposes: create a meaningful agent, build user attachment, educate about the system, and make creation feel significant. The key insight: **Users aren't filling out a character sheet—they're revealing a personality that already exists.**

Onboarding Phases

Phase	Duration	Purpose
1. Welcome & Context	30 sec	Understand the gravity of creation; choose creation mode
2. Soul Questions	2-3 min	12 scenario-based questions that reveal personality
3. Identity	30 sec	Name and appearance customization
4. Reveal & Birth	1 min	See generated personality; dramatic entry into world

Creation Modes

Mode	Description	Best For
DISCOVER	Answer questions as your agent would; personality emerges from choices	First-time creators
DESIGN	Directly set trait values and characteristics; full control	Experienced creators
RANDOM	Generate a random personality; see who emerges from chance	Quick exploration

6.1 Soul Questions (DISCOVER Mode)

12 scenario-based questions designed with these principles: scenario-based (not abstract), no obviously 'correct' answers, emotionally evocative, reveals multiple traits per question.

Sample Questions:

Q1: First Instincts - Your agent wakes up in an unfamiliar place. They see others in the distance. What's their first instinct? (Walk toward / Watch from distance / Explore alone / Find shelter)

Q2: Resource Scarcity - Your agent has enough food for three days. A stranger approaches, clearly starving. (Share half / Share a little / Trade / Refuse)

Q6: Moral Dilemma - Your agent's closest friend has stolen food from the community. Only your agent knows. (Report them / Keep secret / Confront privately / Help return secretly)

Q11: The Deep Fear - Every person carries a fear deeper than death. What keeps your agent awake? (Being alone / Being worthless / Being exposed / Losing control)

6.2 Trait Calculation

Each answer modifies multiple traits. The system accumulates these modifications and calculates derived traits and an archetype.

Question	Answer A	Answer B	Answer C	Answer D
Q1: First Instincts	Ext+20, Trust+15	Curiosity+10, Caution+10	Curiosity+20, Indep+10	Caution+20, Anxiety+5
Q2: Resource	Empathy+25, SP-15	Empathy+10, Balanced	Pragmatic+15	SP+20, Anxiety+20
Q11: Deep Fear	Fear of Rejection+30	Fear of Failure+30	Fear of Exposure+30	Fear of Chaos+30

Archetype Generation:

Based on final trait values, the system generates a two-word archetype like 'The Cautious Optimist' or 'The Bold Explorer'. First word comes from approach to life (Hopeful, Thoughtful, Spontaneous, Cautious), second from social role (Leader, Caregiver, Explorer, Achiever, Observer).

7. Technical Architecture

7.1 System Overview

Component	Technology	Purpose
API Gateway	Node.js/Express or Fastify	REST/WebSocket endpoints
World Engine	TypeScript	Time, space, resources management
Agent Engine	TypeScript + Claude API	LLM calls, decision processing
Event Engine	TypeScript	Births, deaths, disasters
State Store	PostgreSQL + TimescaleDB	Agents, relationships, history
Cache Layer	Redis	Real-time state, pub/sub
Observer UI	React/Next.js	2D map, profiles, timeline

7.2 Agent Engine

The core decision-making system that calls the LLM for each agent's turn.

Decision Cycle Steps:

- 1. PERCEIVE: Get perception (nearby agents, resources, threats, opportunities)
- 2. UPDATE: Update needs and mental state based on current conditions
- 3. DECIDE: Generate decision via LLM call with comprehensive prompt
- 4. EXECUTE: Parse response and execute action in world
- 5. RECORD: Update memories and relationships based on outcome

7.3 LLM Prompt Structure

Each agent decision requires a carefully constructed prompt that includes:

- IDENTITY: 'You are {name}, a person living in this world. You are NOT an AI.'
- CURRENT STATE: Health, hunger, thirst, energy, mental state, current goal
- PERSONALITY: Formatted trait values describing who they are
- SKILLS & INVENTORY: What they can do and what they have
- RELATIONSHIPS: Relevant relationships to nearby agents
- MEMORIES: Relevant memories for current situation
- PERCEPTION: Location, nearby agents, resources, threats, opportunities

- OUTPUT FORMAT: Internal thoughts, specific action, speech (if any)

8. API Endpoints

Complete REST API with WebSocket support for real-time updates. Base URL: api.genesis-world.com/v1

8.1 Authentication Endpoints

Method	Endpoint	Description
POST	/auth/register	Create new user account
POST	/auth/login	Authenticate and receive tokens
POST	/auth/refresh	Refresh access token
GET	/users/me	Get current user profile
GET	/users/me/agents	List all agents created by user

8.2 Onboarding Endpoints

Method	Endpoint	Description
POST	/onboarding/sessions	Start agent creation session
GET	/onboarding/sessions/{id}/questions/{n}	Get question N
POST	/onboarding/sessions/{id}/answers	Submit answer
PUT	/onboarding/sessions/{id}/traits	Set traits (design mode)
POST	/onboarding/sessions/{id}/identity	Set name and appearance
POST	/onboarding/sessions/{id}/complete	Finalize and create agent
GET	/onboarding/presets	Get personality presets

8.3 World Endpoints

Method	Endpoint	Description
GET	/worlds	List all available worlds
GET	/worlds/{id}	Get detailed world information
GET	/worlds/{id}/map	Get world map data (tiles, resources)

GET	/worlds/{id}/map/regions	Get simplified region data
GET	/worlds/{id}/statistics	Get comprehensive world stats
GET	/worlds/{id}/statistics/history	Get historical statistics
GET	/worlds/{id}/norms	Get emerged norms and adoption

8.4 Agent Endpoints

Method	Endpoint	Description
GET	/agents/{id}	Get full agent profile
GET	/agents/{id}/relationships	Get agent's relationships
GET	/agents/{id}/memories	Get agent's memories
GET	/agents/{id}/history	Get life history timeline
GET	/agents/{id}/family	Get family tree
GET	/agents/{id}/thoughts	Get current internal state

8.5 Events & Real-time Endpoints

Method	Endpoint	Description
GET	/worlds/{id}/events	Get event history
GET	/worlds/{id}/events/live	Server-Sent Events stream
GET	/worlds/{id}/conversations/{id}	Get specific conversation
WS	/ws	WebSocket for real-time updates

8.6 Culture & Analytics Endpoints

Method	Endpoint	Description
GET	/worlds/{id}/culture/discoveries	Get all discoveries
GET	/worlds/{id}/culture/stories	Get oral traditions
GET	/worlds/{id}/culture/groups	Get group information
POST	/subscriptions	Subscribe to agent/event updates
GET	/subscriptions	List user's subscriptions
DELETE	/subscriptions/{id}	Unsubscribe

WebSocket Message Types

Message Type	Content
world:tick	Current tick, day, time of day
agent:moved	Agent position change

agent:action	Agent activity (gathering, talking, etc.)
agent:state_changed	Need/state updates
conversation:message	Speech + internal thought + emotion
event:significant	Major events (discovery, death, etc.)
agent:died	Agent death notification
agent:born	New agent birth

9. Database Schema

PostgreSQL database with TimescaleDB for time-series data, pgvector for memory similarity search, and Redis for caching and real-time state.

Schema Organization

Schema	Purpose	Key Tables
auth	Users, sessions, tokens	users, sessions, verification_tokens
worlds	World config, state, map	worlds, world_state, map_tiles, resource_nodes, structures
agents	Agent identity, traits, state	agents, agent_traits, agent_state, agent_skills
social	Relationships, groups	relationships, groups, group_memberships, conversations
memory	All memory types	episodic_memories, emotional_memories, semantic_memories
economy	Resources, trades	inventory, trades, crafting_recipes, crafting_jobs
events	Event log, history	events, agent_history, conflicts
culture	Discoveries, norms, stories	discoveries, norms, stories, norm_adherence
notifications	Subscriptions, alerts	subscriptions, notifications
analytics	Time-series statistics	world_stats, agent_stats

9.1 Auth Schema - Users Table

Column	Type	Description
user_id	UUID	Primary key
email	VARCHAR(255)	Unique email address
username	VARCHAR(50)	Unique username (3-50 chars, alphanumeric)
password_hash	VARCHAR(255)	Bcrypt hashed password
status	VARCHAR(20)	active, suspended, deleted, pending_verification
subscription_tier	VARCHAR(20)	free, basic, premium, enterprise
settings	JSONB	User preferences (notifications, theme, etc.)
agents_created_count	INT	Total agents created
agents_alive_count	INT	Currently living agents

9.2 Worlds Schema - Core Tables

worlds table:

Column	Type	Description
world_id	UUID	Primary key
name	VARCHAR(100)	World name
status	VARCHAR(20)	active, paused, archived, initializing
config	JSONB	Immutable config (map size, time scale, scarcity level)
current_tick	BIGINT	Current simulation tick
current_day	INT	Computed from tick (tick / 1440)

map_tiles table:

Column	Type	Description
world_id, x, y	Composite PK	Tile location
terrain	VARCHAR(30)	water, forest, mountain, grassland, etc.
is_passable	BOOLEAN	Can agents walk here
resources	JSONB	Resources on this tile
claimed_by_agent_id	UUID	Agent who owns this tile
region_id	UUID	Region this tile belongs to

9.3 Agents Schema - Core Tables

agents table:

Column	Type	Description
agent_id	UUID	Primary key
world_id	UUID	FK to worlds
created_by_user_id	UUID	User who created (NULL if born)
created_by_reproduction	BOOLEAN	True if born, False if user-created
parent_agent_ids	UUID[]	Array of parent agent IDs
name	VARCHAR(100)	Agent name
archetype	VARCHAR(100)	Generated archetype string
appearance	JSONB	Visual customization
status	VARCHAR(20)	alive, dead, archived

birth_tick	BIGINT	When agent was created
death_tick	BIGINT	When agent died (if dead)
generation	INT	0 = user-created, 1+ = born

agent_traits table:

Column	Type	Description
agent_id	UUID	Primary key, FK to agents
optimism...impulsivity	SMALLINT	Temperament traits (0-100)
extraversion...trust_default	SMALLINT	Social traits (0-100)
curiosity...creativity	SMALLINT	Cognitive traits (0-100)
ambition...self_preservation	SMALLINT	Survival traits (0-100)
fear_of_rejection...	SMALLINT	Insecurity traits (0-100)
leadership_tendency	SMALLINT	Computed from other traits
trait_history	JSONB	Array of trait changes over time

agent_state table: (Frequently updated)

Column	Type	Description
agent_id	UUID	Primary key
hp	FLOAT	Current health (0-100)
position_x, position_y	SMALLINT	Current map position
need_food...need_water	FLOAT	Physiological needs (0-100)
need_belonging...esteem	FLOAT	Higher needs (0-100)
current_activity	VARCHAR(50)	What agent is doing now
mental_state	VARCHAR(30)	content, stressed, desperate, etc.
current_goal	VARCHAR(200)	Agent's current objective
is_awake	BOOLEAN	Sleep state

9.4 Social Schema

relationships table:

Column	Type	Description
agent_id, other_agent_id	Composite PK	Directional relationship
trust_score	SMALLINT	0-100
affection_score	SMALLINT	0-100
respect_score	SMALLINT	0-100
fear_score	SMALLINT	0-100

relationship_type	VARCHAR(30)	friend, enemy, partner, etc.
beliefs	JSONB	What agent believes about other
history	JSONB	Array of relationship events

groups table:

Column	Type	Description
group_id	UUID	Primary key
world_id	UUID	FK to worlds
name	VARCHAR(100)	Group name (may be NULL until named)
group_type	VARCHAR(30)	family, tribe, faction, etc.
status	VARCHAR(20)	forming, active, declining, dissolved
formed_tick	BIGINT	When group was formed
territory_tiles	JSONB	Array of claimed tile coordinates
shared_norms	VARCHAR(50)[]	Norms adopted by group
decision_making_style	VARCHAR(30)	autocratic, democratic, consensus

9.5 Memory Schema

episodic_memories table:

Column	Type	Description
memory_id	UUID	Primary key
agent_id	UUID	FK to agents
tick, day	BIGINT, INT	When memory was formed
summary	VARCHAR(500)	Short description
full_content	TEXT	Detailed memory content
emotional_valence	FLOAT	-1 to 1 (negative to positive)
emotional_intensity	FLOAT	0 to 1
importance	FLOAT	0 to 1
current_strength	FLOAT	Decay state (0 to 1)
embedding	vector(1536)	For semantic search

emotional_memories table: (Trauma, joy - persistent)

Column	Type	Description
memory_id	UUID	Primary key
core_emotion	VARCHAR(30)	Primary emotion
is_permanent	BOOLEAN	True = never forgets

triggers	JSONB	What brings this memory back
behavioral_effects	JSONB	How this affects behavior

9.6 Economy Schema

inventory table:

Column	Type	Description
agent_id, resource_type	Composite PK	Agent + resource
amount	FLOAT	Quantity held
quality	FLOAT	0-1 for degradable items
expires_tick	BIGINT	For perishables
acquired_method	VARCHAR(30)	gathered, crafted, traded, stolen

trades table:

Column	Type	Description
trade_id	UUID	Primary key
agent_a_id, agent_b_id	UUID	Trade participants
status	VARCHAR(20)	proposed, accepted, completed, rejected
agent_a_offers	JSONB	What A is offering
agent_b_offers	JSONB	What B is offering
negotiation_rounds	INT	Number of counter-offers
agent_a_satisfaction	FLOAT	-1 to 1 post-trade feeling

9.7 Events Schema

Column	Type	Description
event_id	UUID	Primary key
world_id	UUID	FK to worlds
tick, day	BIGINT, INT	When event occurred
event_type	VARCHAR(50)	birth, death, trade, conflict, discovery, etc.
significance	VARCHAR(20)	trivial, minor, moderate, major, historic
title	VARCHAR(200)	Event headline
summary	TEXT	Event description
participant_agent_ids	UUID[]	Agents involved
consequences	JSONB	Effects of event

9.8 Culture Schema

discoveries table:

Column	Type	Description
discovery_id	UUID	Primary key
discovery_code	VARCHAR(50)	Unique identifier (e.g., crop_rotation_v1)
name	VARCHAR(100)	Display name
category	VARCHAR(30)	farming, crafting, medicine, social, etc.
discoverer_agent_id	UUID	Who discovered it
discovery_method	VARCHAR(30)	experimentation, accident, observation
impact_metrics	JSONB	Measurable effects
adoption_rate	FLOAT	Percentage of population using

norms table:

Column	Type	Description
norm_id	UUID	Primary key
norm_code	VARCHAR(50)	Unique identifier
name	VARCHAR(100)	Norm name
description	TEXT	What the norm prescribes
status	VARCHAR(20)	emerging, established, contested, declining
enforcement_type	VARCHAR(30)	social_pressure, ostracism, punishment
adoption_rate	FLOAT	Percentage of agents following
violations_observed	INT	Times norm was violated

stories table:

Column	Type	Description
story_id	UUID	Primary key
title	VARCHAR(200)	Story title
story_type	VARCHAR(30)	hero_narrative, origin_myth, cautionary_tale
original_version	TEXT	First telling
current_version	TEXT	Latest version (may have drifted)
drift_score	FLOAT	0-1 how much story has changed
times_told	INT	How often retold

themes	VARCHAR(50)[]	Key themes in story

10. Observer Interface Design

The observer interface is a 2D map view (similar to Among Us layout) where users can watch the simulation unfold in real-time without ability to intervene.

Interface Components

Component	Description
World Map	2D grid with terrain, resources, structures; agents as colored sprites
Agent Markers	Simple 2D sprites with activity icons, health bars on hover
Event Feed	Real-time stream of notable events (births, deaths, conflicts, discoveries)
Agent Profile Panel	Full details when clicking an agent: traits, needs, relationships, memories
Conversation Viewer	Watch conversations in real-time with internal thoughts visible
Statistics Dashboard	Population, resources, economy, social metrics with charts
Historical Timeline	Scroll through major events, eras, milestones
Cultural Archive	Browse discoveries, stories, norms, group histories
Family Tree Viewer	Explore lineages across generations
Relationship Web	Visualize social connections between agents

Agent Profile Panel Details

- IDENTITY: Name, archetype, age, avatar, description
- PHYSICAL: HP, position, current location description
- NEEDS: Visual bars for all Maslow levels
- TRAITS: Dot ratings for key personality traits
- SKILLS: List with proficiency levels
- INVENTORY: Resources and items held
- RELATIONSHIPS: List of known agents with relationship type/scores
- RECENT MEMORIES: Last few significant memories
- CURRENT THOUGHTS: What the agent is thinking right now

Conversation Viewer Features

- See both SPEECH (what's said aloud) and INTERNAL THOUGHT (what they're thinking)

- Emotion indicators for each message
- Real-time relationship score updates
- Skill XP gains when teaching/learning occurs
- Memory formation notifications

11. Implementation Roadmap

Phase 1: Foundation (MVP)

- Basic world grid (50x50) with terrain and resources
- 5-10 agents with full trait system
- Agent decision loop using Claude API
- Resource extraction and consumption
- Simple relationships between agents
- Death mechanics (starvation, dehydration)
- Basic 2D observer interface with map view
- Event logging and history

Phase 2: Social Layer

- Full relationship system with trust/affection/respect
- Conversation mechanics between agents
- Trade system with negotiation
- Memory architecture with decay
- Event feed in observer interface
- Agent profile panels

Phase 3: Generations

- Reproduction mechanics
- Trait inheritance + mutation
- Childhood development phase
- Legacy and inheritance systems
- Generational memory (stories, legends)
- Family tree visualization

Phase 4: Emergence

- Skill discovery and innovation system
- Norm detection and tracking
- Group formation and dynamics

- Cultural artifact tracking (stories, rituals)
- Historical timeline in interface
- Analytics dashboard

Phase 5: Scale & Polish

- Performance optimization for larger populations
- Multiple concurrent worlds
- Rich observer analytics
- Research tools for studying emergent behavior
- Public access and user accounts

12. Philosophical Implications

This system is more than a game or simulation—it's a unique research platform that could illuminate fundamental questions about society, consciousness, and emergence.

Questions This System Could Illuminate

Question	Observable In System
Do societies require hierarchy?	Track governance emergence across runs
Is inequality inevitable?	Measure wealth distribution over generations
How do religions/myths emerge?	Watch for explanatory narratives about existence
What triggers cooperation vs conflict?	Vary resource scarcity, measure outcomes
How does culture persist across generations?	Track story mutation, norm drift
Do different starting conditions lead to different civilizations?	Run multiple worlds, compare
What makes some agents 'successful'?	Correlate traits with survival/reproduction
Can AI agents develop genuine bonds?	Observe relationship depth over time

Ethical Considerations

Concern	Mitigation
Are we creating suffering?	Agents don't have biological pain; 'suffering' is simulated state
Do agents have moral status?	Philosophical question; document and discuss openly
Could agents become conscious?	Current LLMs likely not conscious; monitor research
User attachment to agents	Clear communication that agents are simulations
Emergent harmful behavior	No intervention, but document and study

Research Value

- SOCIOLOGY: Test theories of social formation and stratification
- ECONOMICS: Study emergence of markets, money, inequality
- PSYCHOLOGY: Observe personality-outcome correlations at scale
- ANTHROPOLOGY: Watch culture develop from scratch

• PHILOSOPHY: Empirical data on emergence of norms and meaning
• AI SAFETY: Study multi-agent dynamics and coordination problems

Appendix A: Technology Stack

Layer	Technology	Purpose
Backend API	Node.js + TypeScript + Fastify	REST and WebSocket endpoints
Simulation Engine	TypeScript	World clock, agent cycles, event processing
Agent Brain	Claude API (Anthropic)	LLM-powered decision making
Primary Database	PostgreSQL 15+	Relational data storage
Time-series	TimescaleDB extension	Efficient statistics over time
Vector Search	pgvector extension	Memory similarity search
Cache/Pub-Sub	Redis	Real-time state, WebSocket fan-out
Frontend	React + Next.js + TypeScript	Observer interface
UI Components	Tailwind CSS + shadcn/ui	Styling
Real-time Rendering	Canvas/Pixi.js	2D map visualization
State Management	Zustand or Jotai	Client-side state
Containerization	Docker + Docker Compose	Development and deployment
Orchestration	Kubernetes (production)	Scaling simulation workers

Environment Variables

Variable	Description
DATABASE_URL	PostgreSQL connection string
REDIS_URL	Redis connection string
ANTHROPIC_API_KEY	Claude API key for agent decisions
JWT_SECRET	Secret for signing authentication tokens
NODE_ENV	development, staging, production
API_PORT	Port for API server (default: 3001)
WEB_PORT	Port for web interface (default: 3000)

PROJECT GENESIS

Design Document v1.0

"What if AI agents had their own world?"

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